

2d Produce the linear and quadratic taylor polynomials for $f(x) = \ln(1 + e^x)$, $a = 0$. Then Graph in Matlab.

$$\text{Linear: } p_1(x) = f(a) + (x - a)f'(a)$$

$$p_1(x) = 1 + \frac{1}{2}x$$

$$\text{Quad: } p_2(x) = f(a) + (x - a)f'(a) + \frac{1}{2}(x - a)^2 f''(a)$$

$$p_2(x) = 1 + \frac{1}{2}x + \frac{1}{8}x^2$$

3c Produce a general formula for the degree N Taylor polynomial for the following function

$$f(x) = \sqrt{1+x}$$

$$p_n = \sum_{j=0}^n \frac{(x-a)^j}{j!} f^{(j)}(a)$$

$$p_n(x) = 1 + \binom{1/2}{1}x + \binom{1/2}{2}x^2 + \dots + \binom{1/2}{n}x^n$$

2 Find the degree 2 taylor polynomial appx to $f(x) = e^x \sin(x)$, $a = 0$. Bound the error in this appox when $-\frac{\pi}{4} \leq x \leq \frac{\pi}{4}$.

$$f^1(x) = e^x(\sin(x) + \cos(x)), f^1(a) = 1$$

$$f^2(x) = 2e^x \cos(x), f^2(a) = 2$$

$$p_2 = \left(\frac{x^0}{1} * 0\right) + \left(\frac{x^1}{1} * 1\right) + \left(\frac{x^2}{2} * 2\right)$$

$$p_2 = x + x^2$$

5 How large should the degree $2n-1$ be chosen in 1.14 to have $|\sin(x) - p_{2n-1}(x)| \leq .001$. For all $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$. Check by evaluating the resulting $p_{2n-1}(x)$ at $x = \frac{\pi}{2}$

$$N = 7$$